

ST512-SP26-Homework-1

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Problem 1

$$n = 36$$

$$\bar{x} = 23.4$$

$$sd(s) = 7.2$$

$$CI: \bar{x} \pm t^* \times SE$$

$$SE = \frac{s}{\sqrt{n}}$$

$$SE = \frac{7.2}{\sqrt{36}} = 1.2$$

$$t^* = qt(1-(.04/2), 36) = 2.13$$

$$CI: 23.4 \pm 2.13 * 1.2$$

$$96\% CI: (20.84, 25.96)$$

Problem 2

$$H_o : \mu = 3.5$$

$$H_A : \mu < 3.5$$

Will be conducting a left-tailed test because we want to know if the drug delivers relief faster (i.e., less than 3.5 minutes).

$$n = 50$$

$$\bar{x} = 3.1$$

$$s = 1.5$$

$$\alpha = 0.05$$

$$SE = \frac{s}{\sqrt{n}} = \frac{1.5}{\sqrt{50}} = 0.212$$

$$t\text{-stat} = \frac{\bar{x} - \mu_o}{SE} = \frac{-0.4}{0.212} = -1.887$$

$$p\text{-value} = \text{pt}(t\text{-value}, df) = 0.033$$

$p\text{-value} < \alpha$, we reject H_o

Therefore, there is sufficient evidence at the 5% confidence interval to conclude that the newly developed pain reliever delivers perceptible pain relief more quickly than the standard pain reliever.

Problem 3

	Design 1	Design 2
n_i	11	6
$\bar{y}_{i\cdot}$	52	46
sd	12	10

a

$$\text{point estimate: } \mu_1 - \mu_2 = \bar{y}_{1\cdot} - \bar{y}_{2\cdot} = 52 - 46 = 6$$

$$95\% \text{ CI: } (\bar{y}_{1\cdot} - \bar{y}_{2\cdot}) \pm t^* \times SE$$

Assuming equal variance:

$$df = n_1 + n_2 - 2 = 15$$

$$s_p^2 = \frac{(n_1-1)s_1^2 + (n_2-1)s_2^2}{df} = \frac{10(12^2) + 5(10^2)}{15} = 129.33$$

$$s_p = \sqrt{129.33} = 11.37$$

$$SE = s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}} = 11.38 \sqrt{\frac{1}{11} + \frac{1}{6}} = 5.78$$

$$t^* = \text{qt}(0.975, 15) = 2.13$$

$$95\% \text{ CI: } 6 \pm 2.13 * 5.78$$

$$95\% \text{ CI: } (-6.31, 18.31)$$

b

$$H_o : \mu_1 = \mu_2$$

$$H_A : \mu_1 \neq \mu_2$$

$$t\text{-stat} = \frac{\text{pointestimate}-0}{SE} = \frac{6}{5.78} = 1.04$$

$$p\text{-value: } 2 * p(T > |t\text{-value}|) = 0.31$$

$$\alpha = 0.01$$

p-value $> \alpha$; fail to reject H_o

There is not sufficient evidence at the 1% significance level to conclude the mean monthly sale differ between the two designs.

Problem 4

a

This is an experimental design because the treatments are being controlled, the investigator's role is active

b

This is a randomized complete block design.

c

- i. response variable: tomato yield
- ii. factor: fertilizer application rate
- iii. treatments: 3, 5, and 7 units/acre

d

There are 6 replications per treatment.

Problem 5

a

Source	df	Sum of Squares	Mean Squares
Between	2	1533	766.6
Within (Error)	20	6241	312.0
Total	22	7774	

b

Refer to R code at end for these values:

$$F\text{-value} = 2.457$$

$$p\text{-value} = 0.111$$

Our null hypothesis is that the mean phosphorous content does not differ among the three varieties. Using the calculated F-value, 2.457, the p-value was calculated to be 0.111. In comparison to a standard significance level of 5%, our p-value is larger. Therefore, failing to reject the null hypothesis. I think the three varieties have the same mean phosphorous content.

Problem 6

<i>Source of Variation</i>	<i>df</i>	<i>SS</i>	<i>MS</i>	<i>F-value</i>	<i>P-value</i>
Between	4	2.281	0.570	4.32	0.0039
With-in (error)	60	7.949	0.132		
Total	64	10.230			

Based on the ANOVA table, the calculated p-value is approximately 0.0039. The p-value is less than the significance level of 0.01, thus we reject the null hypothesis. Therefore, it appears to be a difference in means between the five diets.

R Code

```
qt(1-(.04/2), 36) #problem 1
```

```
[1] 2.130871
```

```
pt(-1.887,49) #problem 2
```

```
[1] 0.03254595
```

```
qt(0.975,15) #problem 3.a
```

```
[1] 2.13145
```

```
2*(1-pt(1.04,15)) #problem 3.b
```

```
[1] 0.314814
```

```
# problem 5 r-code
group <- factor(c(rep("A",8), rep("B",8), rep("C",7)))

value <- c(
  rep(72 + sqrt((8-1)*312/8), 4), rep(72 - sqrt((8-1)*312/8), 4),
  rep(68 + sqrt((8-1)*310/8), 4), rep(68 - sqrt((8-1)*310/8), 4),
  rep(55 + sqrt((7-1)*321/7), 3), rep(55 - sqrt((7-1)*321/7), 4)
)

data <- data.frame(group, value)
anova_model <- aov(value ~ group, data = data)
summary(anova_model)
```

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
group	2	1533	766.6	2.457	0.111
Residuals	20	6241	312.0		

```
#problem 6 p-value
1-pf(4.32,4,60)
```

```
[1] 0.003885707
```